

alterspec audit, 2026-10-09

Scope: the whole repository at commit 0365d2f (Phase 8, version 0.5.0). Every source file under `src/`, every shipped asset (templates, prompts, skill and agent wrappers, mockup kit, prototype runtime), the tests, the valid fixture, the Product Catalog example, README, CLAUDE.md and HANDOVER.md were read. Suspected defects were reproduced on copies of the fixture before anything was changed, and the example mockup app was exercised in a browser.

Nothing was committed. All changes are in the working tree for review.

Result

Check	Before	After
npm test	326 yes	338 yes (12 regression tests added)
npm run typecheck, lint, prettier	yes	yes
validate on fixture and example	0	0
views --check on fixture and example	current	current
Example handoff reproduced byte-for-byte	yes	yes (no renderer changed)
Mockup app in the browser (example)	-	sign-in, validation, create, approve: no console errors

Defects found and fixed

Each item was reproduced first; the file in brackets is where the fix lives.

- 1 `alterspec show UX-SCR-...` **failed with "not found"**. Experience screens were missing from the lookup, so the `/alterspec-experience` and reviewer prompts couldn't inspect one. Now found, with their template sections (Layout, Interactions, ...) reported like a capability's. [`src/commands/show.ts`, `src/spec/model.ts`]
- 2 `apply` **could leave a direct-edit error behind**. The baseline was refreshed only for objects the impact counted as modified. An overlay edit the impact ignores (whitespace, a comment) was still written to `spec/`, so the next `validate` reported it as a direct edit. Every object the change touched is now baselined. [`src/changes/apply.ts`]
- 3 `experience sync` **said "aligned" while leaving errors**. When a role was added to a business screen, `sync` never gave it a default view (it only reused the drafted `default` id, which was taken), and it never updated `data-roles` on existing groups and actions whose audience became narrower or wider. Result: `experience-states` and `experience-mockup` errors that the designer had to fix by hand. Sync now adds a `default-<role>` state for every role nobody is shown as, rewrites `data-roles` where the audience changed (and drops it where the element is no longer narrowed), updates the roles on `<body>`, and reports what it changed. [`src/commands/experience.ts`, new `withAttr` in `src/experience/html.ts`]
- 4 **Fresh `experience init` produced an app that couldn't load**. The pages read `_generated/experience/spec.js`, which only `views` writes; after `init` it didn't exist until someone ran `views`. `init` now runs `views`. Inside a change proposal the pages pointed at `changes/CHG-NNN/spec/_generated/experience/spec.js`, which never existed, so a screen designed in a change could not be opened in a browser before `apply`. The experience commands now write a preview copy there (generated output: never merged, never fingerprinted). [`src/commands/experience.ts`]
- 5 **Template examples burned the first IDs**. The screen template's comments rendered to `CAP-<MOD>-001` and `SCR-GLB-01`; new scans every file for the highest used number, so a fresh project's first capability became `CAP-HR-002` and the first shared screen `SCR-GLB-02`. The examples now read `CAP-<MOD>-NNN`, `SCR-GLB-NN`, `SCR-XX-NN`. [`assets/templates/screen.md`, `assets/templates/experience-screen.md`]

- 6 **Template overrides were documented but never used.** README and `.alterspec/custom/README.md` promised that `.alterspec/custom/templates/<name>.md` replaces a template; the CLI always read the shipped one. Templates are now resolved through one function that prefers the project's override: `new`, `change new`, `experience new`, the empty collection file a change creates, and the `incomplete-section` check (which otherwise would demand the shipped headings). `doctor` reports an override whose name matches no shipped template, as it already did for prompts. `[src/lib/templates.ts (new), src/commands/new.ts, src/commands/change.ts, src/commands/experience.ts, src/changes/merge.ts, src/spec/sections.ts, src/lint/*, src/commands/doctor.ts]`
- 7 **Removing an experience screen in a change left its mockup page behind.** `change edit UX-...` copies the mockup along; `change remove UX-...` didn't remove it, leaving an orphan page in `spec/`. It does now. `[src/commands/change.ts]`
- 8 **Small CLI inconsistencies.** The `experience` subcommands printed `spec/...` even with `--spec <other>`; the `--kit` help claimed to refresh `spec.js` (it is views that does). `[src/cli.ts]`
- 9 `experience rebuild` **overwrote hand-designed pages** (requested after the first pass). The contract now records the fingerprint of the page as the CLI last rendered it (`page:`). `rebuild` replaces the page only while it is still that render; otherwise it keeps the page, writes the fresh render to `_generated/experience/rebuild/<SCR>.html` and reports: `data-src` values the business spec lacks (to be taken through the spec change flow), check findings, and whether the kit markup is outdated. `--force` is the only way to overwrite, and the prompts forbid the model from using it. `[src/commands/experience.ts, src/experience/scaffold.ts, src/schemas/experience.ts, src/cli.ts, assets/prompts/experience.md, assets/prompts/agents/ux-designer.md]`

Documentation corrected

- README: four agents, not two (the UX designer and experience reviewer were missing); the example has three applied changes and six screens; template overrides explained with their limits.
- `assets/prompts/experience.md`: what `init` writes and how `spec.js` appears.
- `HANDOVER.md`: a note that the `alter-` prefix and `gray-matter` were superseded (the brief is history, `CLAUDE.md` is current).
- `CLAUDE.md`: templates are read through `src/lib/templates.ts`.

Checked and found sound

These were reviewed closely because they carry the framework's guarantees; no change was needed.

- **IDs:** one source of formats; `new` never reuses a number, including numbers that only appear in archived changes or generated output.
- **Fingerprints and conflicts:** `GENERATED` blocks are reduced to bare markers before hashing, so regenerating views never looks like an edit; a change whose base object moved on is reported as a conflict and `--rebase` is the only way past it.
- **Approval:** `approved_hash` covers overlay files, removals and the proposal body; `apply` refuses anything edited after approval, anything with conflicts, and anything that would leave a lint error.
- **Zero deviation:** the mockup checks read `data-src`, `data-roles`, `data-show-in` and the visible text; the dry hash covers exactly what a screen realises; `handoff` refuses any screen whose experience isn't ready, reviewed and finding-free, with no override.
- **Wrappers:** the 17 skills and 4 agents are thin, prefixed, point at the prompts, and only `validate`, `views` and `impact` may be invoked by the model without the user typing them.
- **Binary assets** survive fingerprinting, overlays, `apply` and `handoff` byte for byte (existing test).

Known limitations left as they are

Not bugs, but worth knowing. None blocks normal use.

- The impact report compares front-matter fields and `##` sections; a change to the text above the first section (the title line) is applied and baselined correctly but not listed as a difference.
- A rejected change stays in `spec/changes/` (only applied changes are archived). It can't be edited or used by `new --change`; delete the folder when it is no longer wanted.
- Template overrides must keep the original front-matter placeholders and GENERATED blocks; the CLI doesn't check that.
- `experience init --kit` refreshes the kit files and the sign-in page only; a hand-designed page built with an older kit is brought over by merging from the reference render `rebuild` writes, not by re-rendering it.

How to verify

```
npm run build && npm test && npm run typecheck && npm run lint && npx prettier --check .  
node dist/cli.js validate examples/catalog && node dist/cli.js views --check examples/catalog
```

Suggested follow-ups (not done, your call)

- Bump to 0.5.1 and regenerate the example handoff (the manifest carries the version), then publish.
- Commit the working tree; the diff is 30 files, all reviewed above.